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WebSphere v9 Administration: Module 9

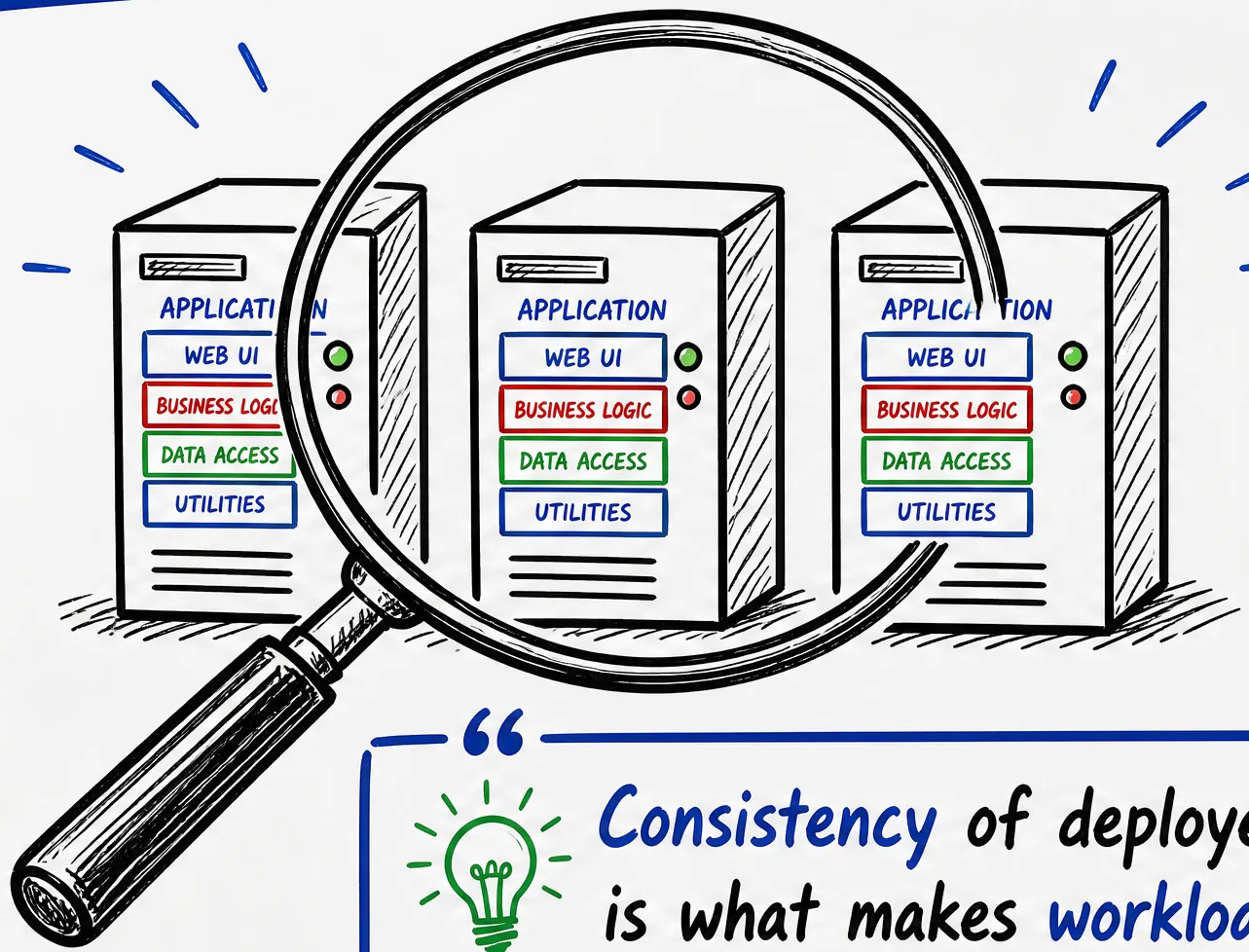
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Static Clusters



> High availability begins the moment you remove the assumption that one process must always be alive.

The Cluster Mental Model



Identical Application Components

> **Consistency** of deployed application components is what makes **workload mobility** safely possible.

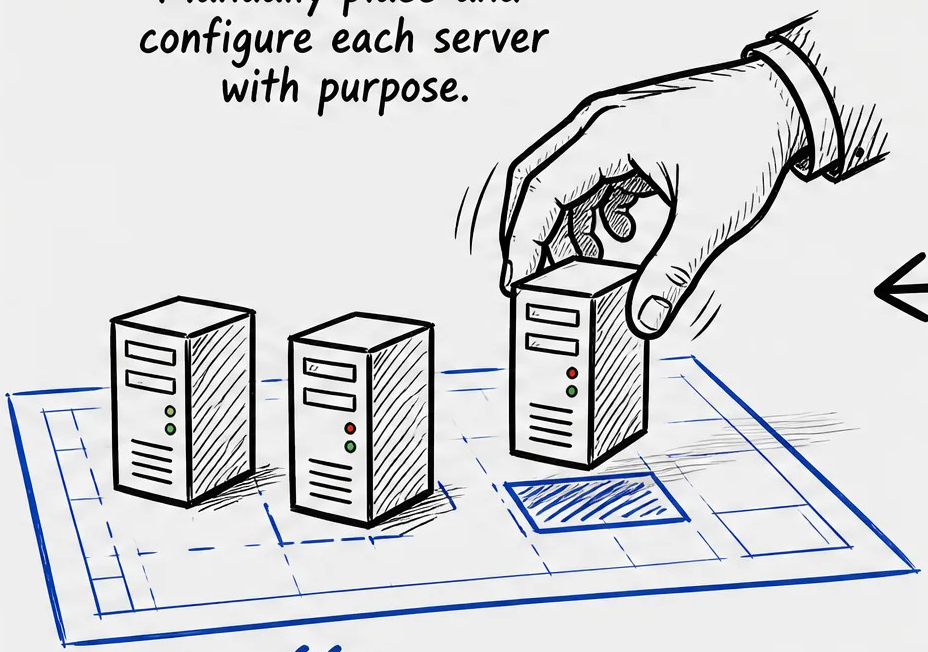
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Why "Static" Matters

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STATIC

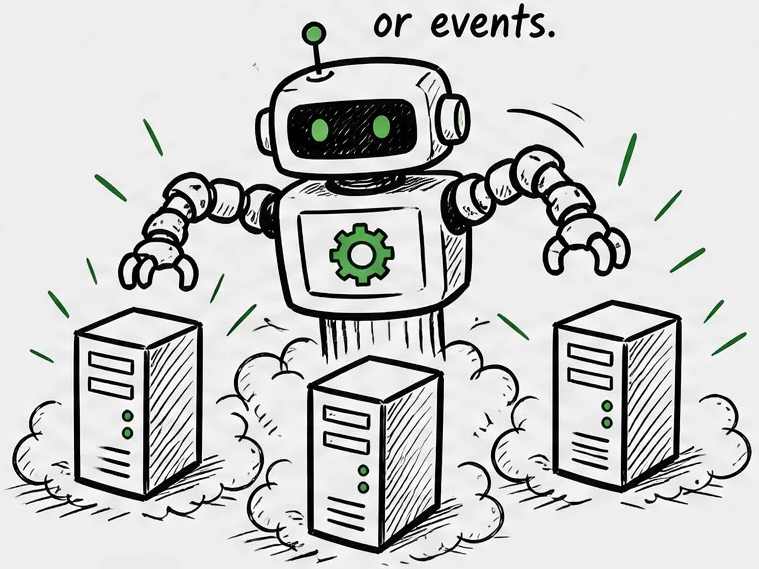
Manually place and configure each server with purpose.



Intentional Administrative Control

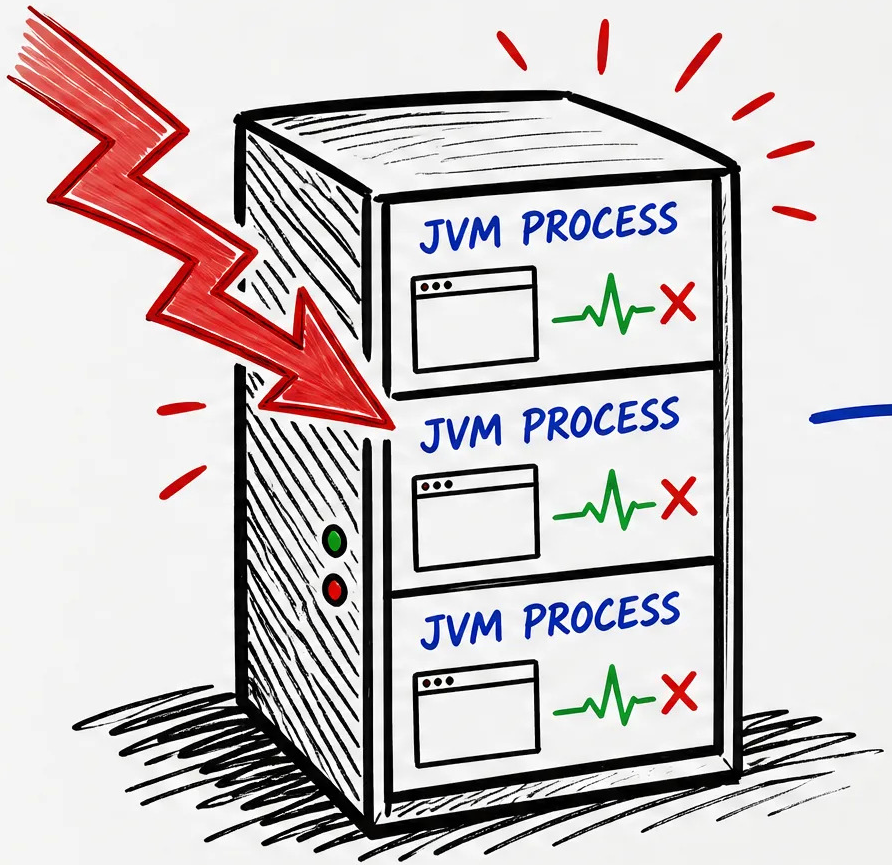
DYNAMIC

Automatically spawn servers on demand based on rules or events.



> A static cluster is ultimately only as consistent as the rigorous configuration discipline behind it.

Vertical Clustering



Isolating Processes, Not Hosts

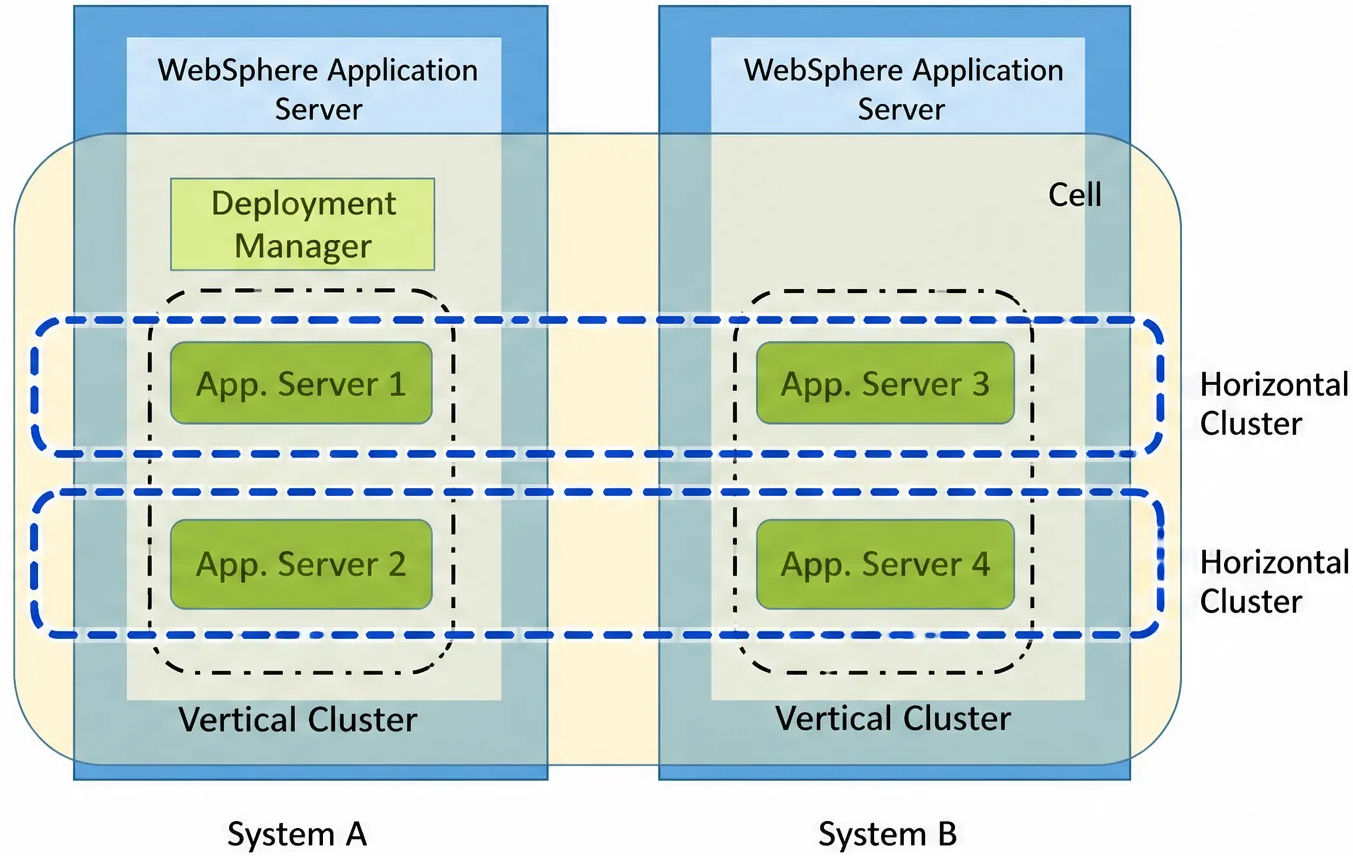
> Separate *software processes* are *not* at all the same thing as separate *hardware failure domains*.

Horizontal Clustering

Surviving Host Failure

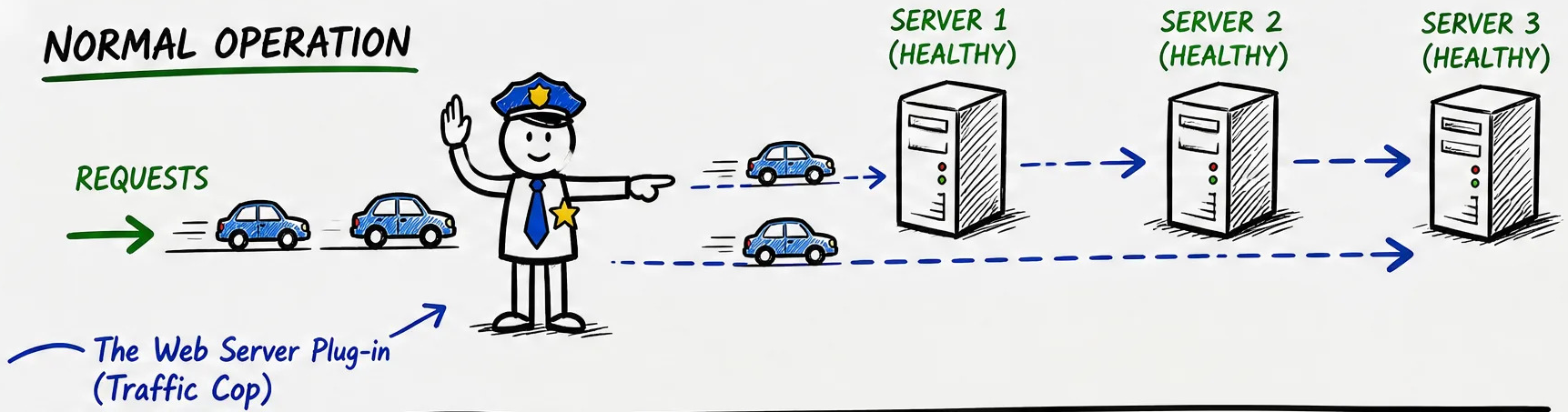


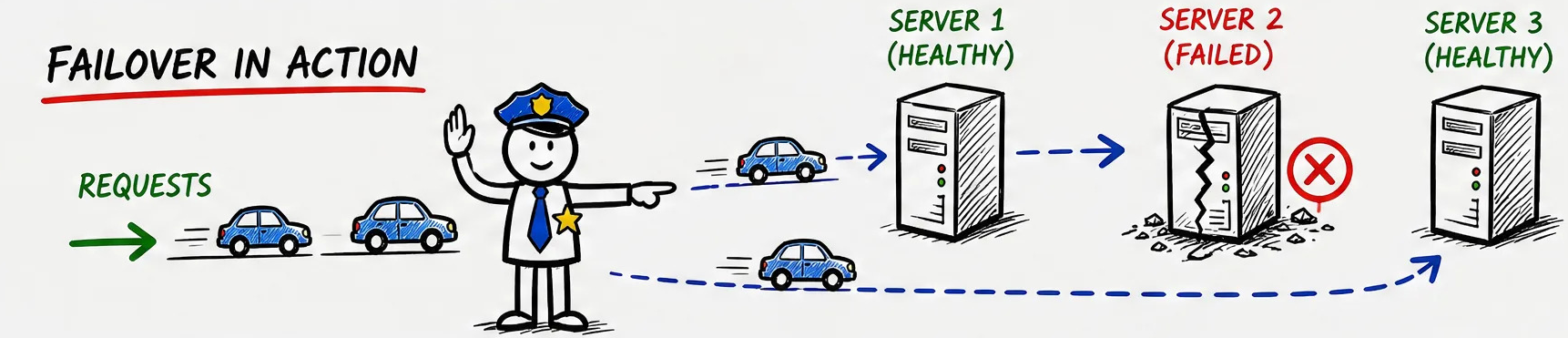
Horizontal placement decisively turns a catastrophic host failure from an outage into a routine workload-routing event.



Workload Management and Failover

The Intelligent Traffic Director

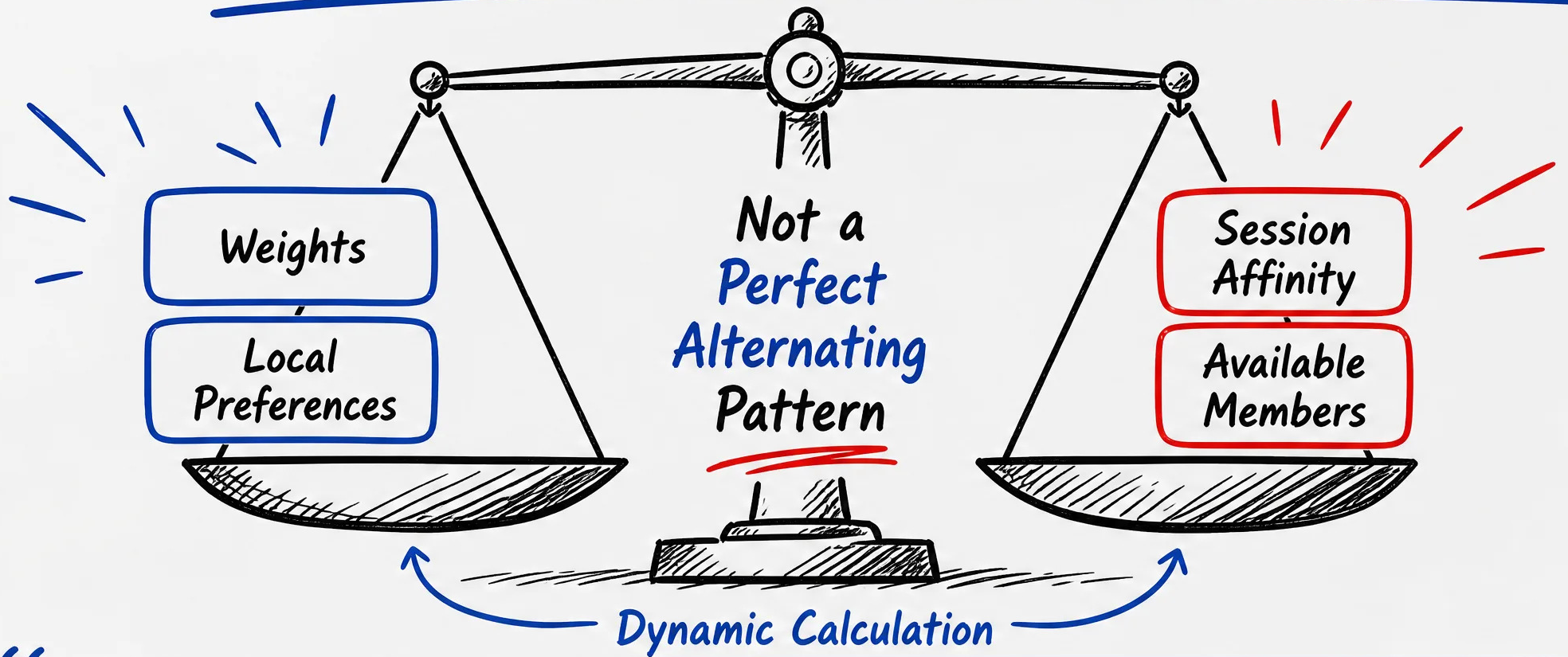




> **Failover** successfully protects the route to healthy members; internal application state still desperately needs its own **resilience strategy**.

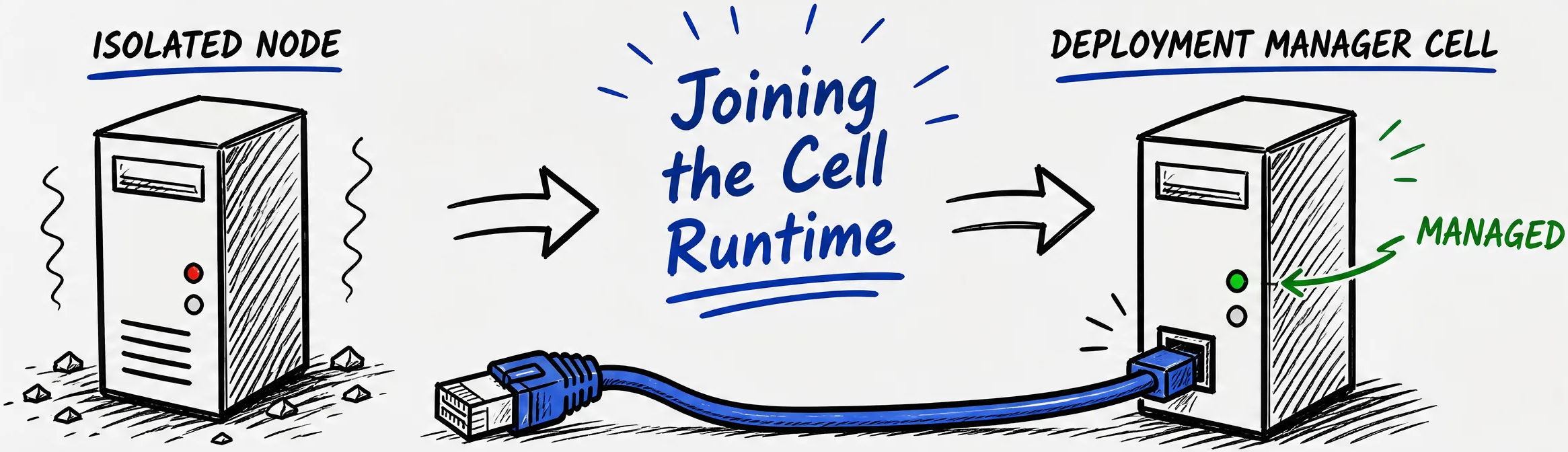
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Request Routing Is a Policy Decision



> Load balancing is an intelligent, policy-driven approximation, not a firm promise of perfectly alternating requests.

Federate the Node Before the Cluster

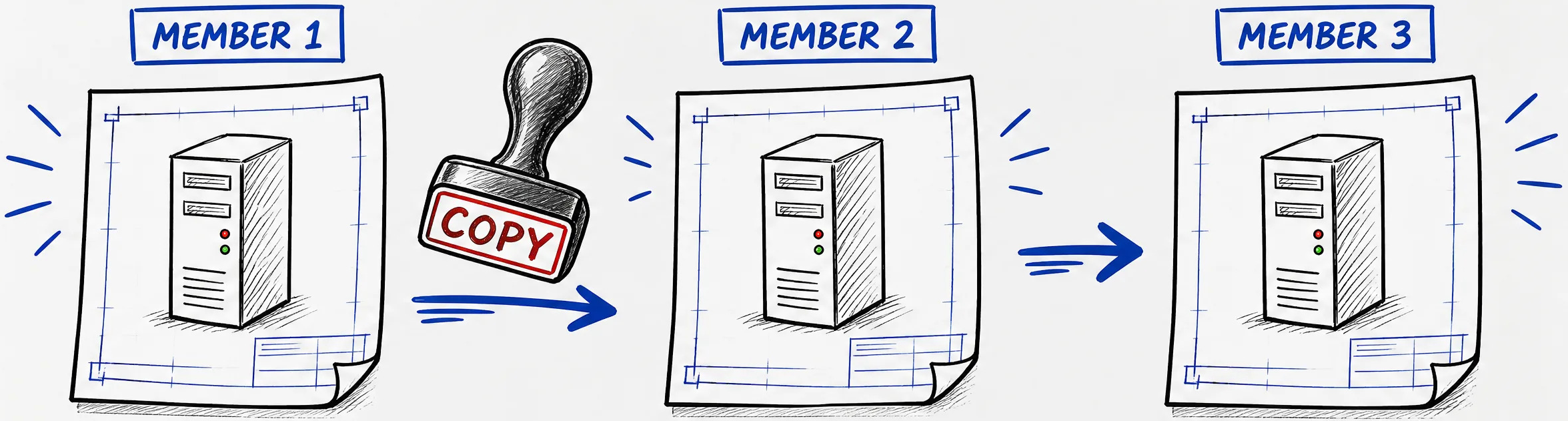


> Federation actively converts an isolated runtime into a highly managed participant in the enterprise cell.

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Create the Static Cluster

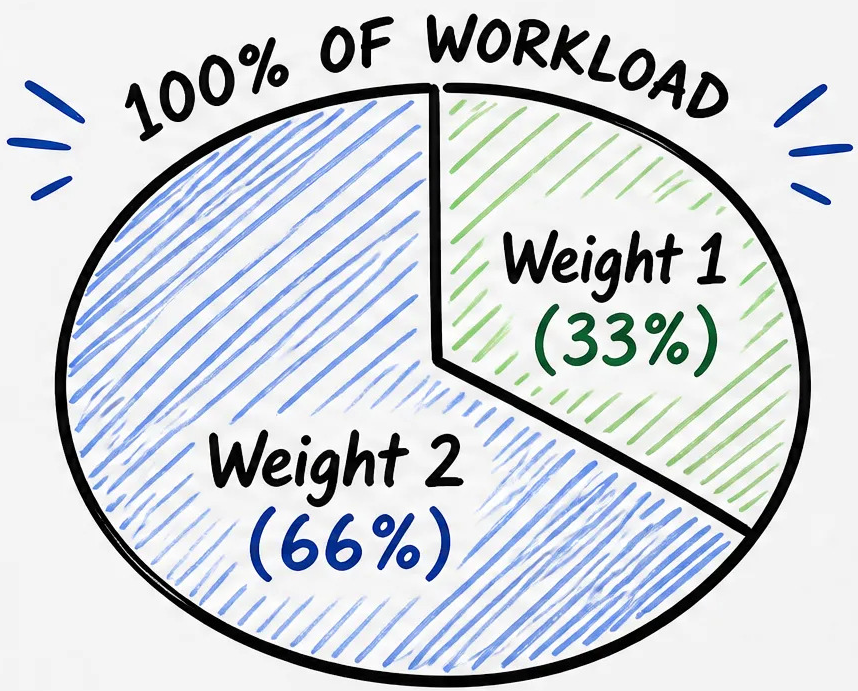
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THE FIRST MEMBER IS THE TEMPLATE

> The first member is not just server number one;
> it becomes the exact pattern for the servers that follow.

Member Weights Are Relative Capacity



A RATIO,
NOT AN
ABSOLUTE
MEASURE



WEIGHT is a **STEERING CONTROL**,
not a **SUBSTITUTE** for **CAPACITY TESTING**.

Verify the Cluster Before Production

STRICT GATES

1. Synced Configuration ☑
2. Identical App Deployment ☑
3. End-to-End Routing Test ☑

→ **Proving Fault Tolerance** →

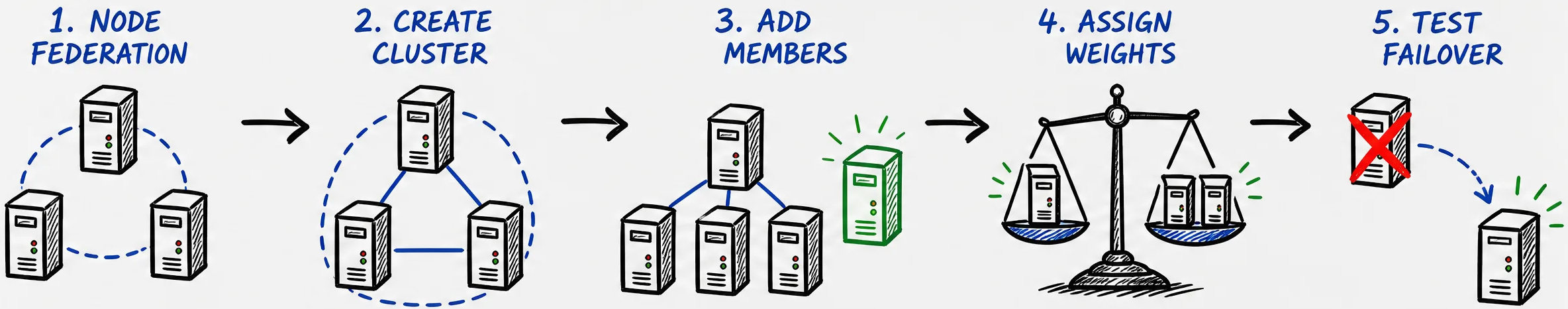


> Failover that has never been tested is merely a theoretical claim, not a real operational capability.

Module 9 Summary and Lab

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Building the Static Cluster



> **A resilient cluster** succeeds only when absolutely any planned member can safely carry the application's full work.